

Test Points

Test point	Details
TP0	0V
TP1	5V
TP2	PWM signals for servo motor (M1)

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno board
- Shield1 - Arduino Ethernet shield
- LED1 - 5mm LED

Resistor (1/4-watt, $\pm 5\%$ carbon):

- R1 - 680-ohm

Miscellaneous:

- M1 - 5V servo motor

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

an Arduino board using long-lead stackable headers (refer Fig. 4). This keeps the pin layout intact and allows another shield to be stacked on top, if needed.

The Ethernet shield has a standard RJ-45 connection, with an integrated line transformer and Power over Ethernet (PoE) enabled.

There is an onboard microSD card slot, which can be used to store files for serving over the network. It is compatible with the Arduino UNO and Arduino Mega (using the Ethernet library). The onboard microSD card reader is accessible through the SD Library. When working with this library, digital pin 4 of the Arduino cannot be used.

The shield also includes a reset controller, to ensure that the W5100 Ethernet module is properly reset on power-up. Previous revisions of the shield are not compatible with the Mega board and need to be manually reset after power-up.

Arduino communicates with both the W5100 and SD card using the SPI bus (through the ICSP header). They are connected to digital pins 10, 11, 12 and 13 on the UNO and pins 50, 51 and 52 on the Mega. On both the boards, pin 10 (slave select or SS pin) is used to select the W5100 and pin 4 for the

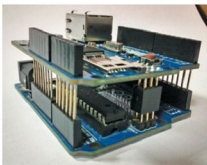


Fig. 4: Ethernet shield on top of Arduino

SD card.

On the Mega, the hardware SS pin (pin 53) is not used to select either W5100 or SD card, but it must be kept as an output, otherwise SPI interface won't work.

Note that, because the W5100 and SD card share the SPI bus, only one can be made active at a time. If you are using both peripherals in your program, this should be taken care of by the corresponding libraries. If you are not using one of the peripherals in your program, however, you will need to explicitly deselect it. To do this with the SD card, set pin 4 as an output and write a 'high' to it. For the W5100, set digital pin 10 as a high output.

The reset button on the shield resets both the W5100 and the Arduino board.

Digital pins 4 and 7 of the shield (Shield1) are connected to LED1 and control pin of servo motor (M1), respectively.

Software

The software for this project is written in Arduino programming language. The Arduino UNO is programmed using Arduino IDE software. ATmega328 on Arduino UNO comes with a boot loader that allows you to upload new code to it without the use of external hardware programmer. It communicates using the STK500 protocol. You can also bypass the boot loader and program the microcontroller through ICSP (in-circuit serial programming) header, but using boot loader programming is quick and easy.

Select the correct board from 'Tools -> Board' menu in Arduino IDE and

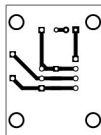


Fig. 5: An actual-size PCB for the controller

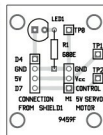


Fig. 6: Component layout for the PCB

burn the program (sketch) through standard USB port in the computer.

Construction and testing

An actual-size, single-side PCB for the web-based device controller using Arduino is shown in Fig. 5 and its component layout in Fig. 6. Assemble the components on the recommended PCB to minimize assembly errors. Carefully assemble the components and double-check for any overlooked error.

To check working of the project, follow the steps below:

1. Connect the servo as shown in the circuit diagram (Fig. 1)
2. Power up the Arduino board through power adaptor
3. Connect Ethernet shield on top of the Arduino board
4. Connect the Ethernet shield with LAN cable through your router
5. Ping the IP address 192.168.0.160, or change the IP address as per your router setting.

You need to change the IP address in the source code, recompile and burn into the microcontroller

Now you can access the servo (M1) and LED1 using this IP address on your browser. If the connection is successful, you can see following user-interface controls on your browser:

1. Turn on LED, Turn off LED
2. Rotate Left, Rotate Right

You can switch on/off LED1 and also control the servo motor M1 by clicking the appropriate button.

To test the circuit for proper functioning, verify correct 5V supply for the circuit at TP1 with respect to TP0. ●

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